

GroundWalks (WalkSensePlace)

Reading the City Through Your Feet
to Design Healthier Urban Places

Kristian **Kloeckl** (CAMD, urban interaction design)
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Amy **Lu** (CAMD/Bouv  , health tech and communication)
RAs Catherine **Zhou**, Samuel **Greeman**
Stappone (technology partner)



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Reading the City Through Your Feet
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"Walking is man's
best medicine."

— Hippocrates

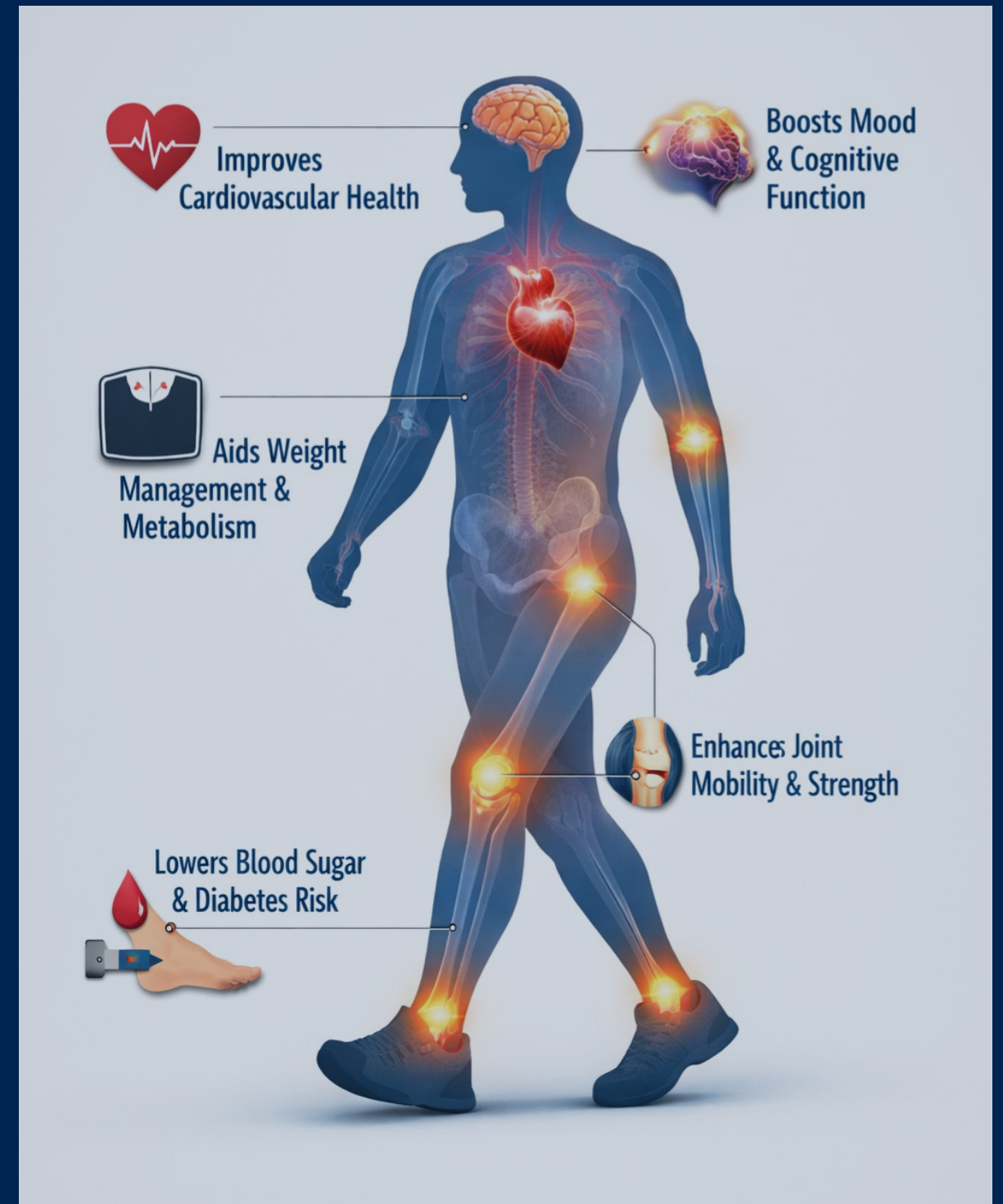
"The rhythm of walking
generates a kind of
rhythm of thinking, ..."

— Rebecca Solnit, *Wanderlust*



Walking...

One of the most **accessible and sustainable forms of physical activity**.
No specialized equipment and minimal training required.



Walkable Cities Movement & Public Health

Walking can be integrated into daily routines.

Even moderate increases in walking can yield significant health gains at the population level.



Walk Score®

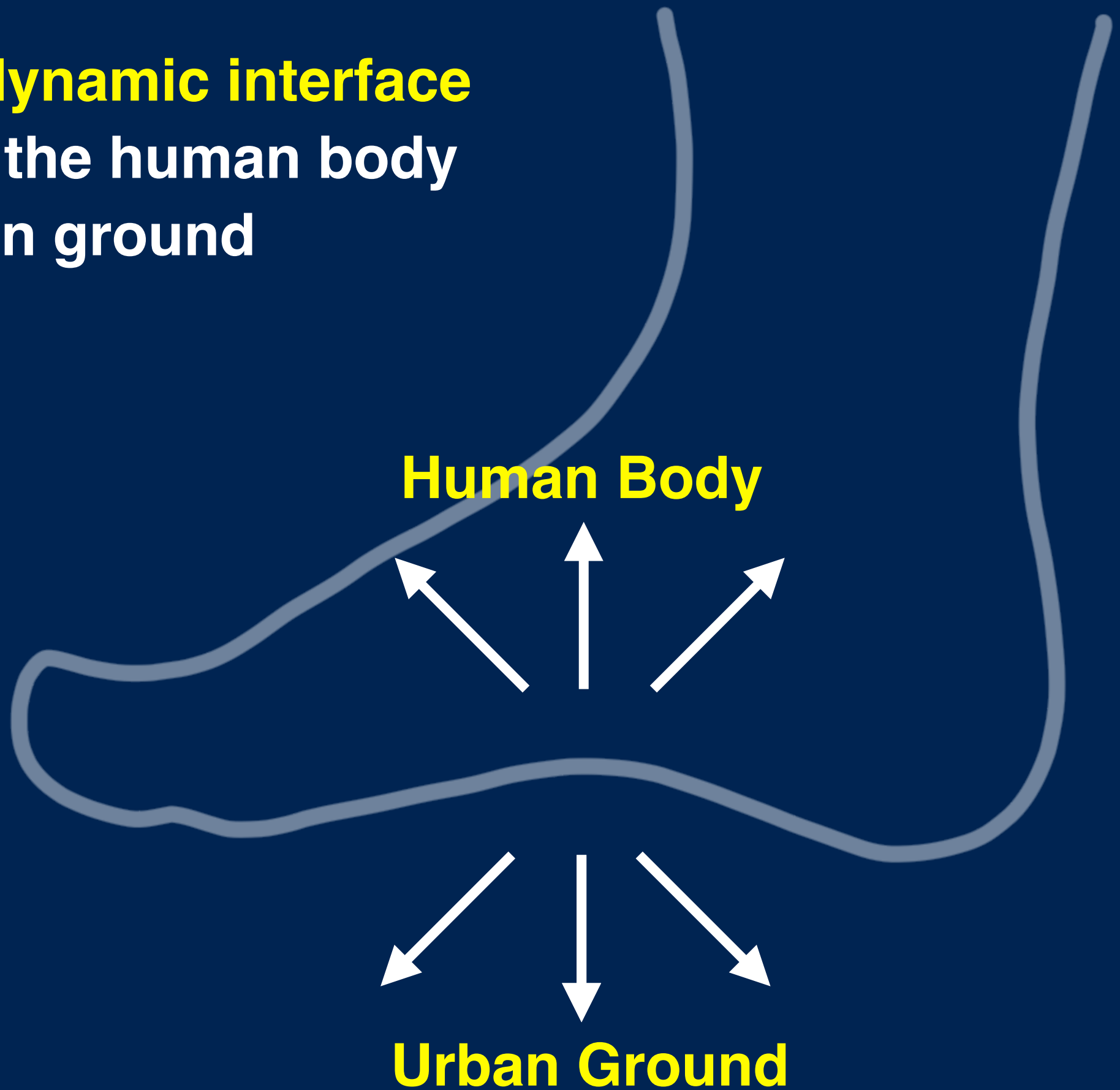


PEDESTRIANS FIRST

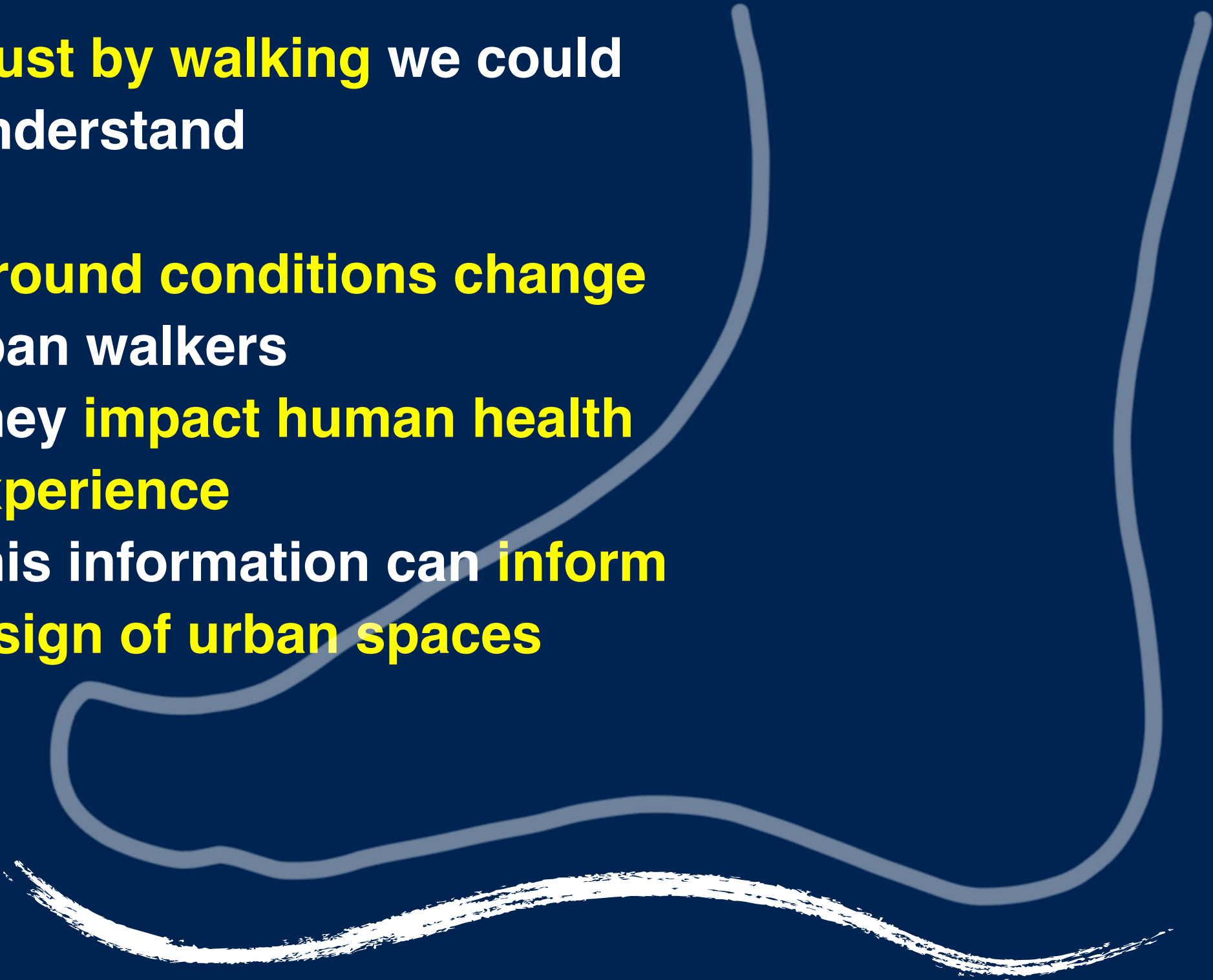
A MIX OF INFRASTRUCTURE, ACTIVITY, AND PRIORITY

INFRASTRUCTURE	ACTIVITY	PRIORITY
ENSURE A PHYSICAL SPACE AND DESIGN THAT PROMOTES WALKING	BRING PEOPLE AND ACTIVITIES CLOSE ENOUGH TO WALK IN SAFE AND LIVELY ENVIRONMENTS	GIVE PREFERENCE TO WALKING, CYCLING, AND TRANSIT OVER PRIVATE CARS
1 Sidewalks are sufficiently wide, in good condition, clean, unobstructed, and protected 2 Crosswalks are accessible for all pedestrians, safe to cross, and sufficiently wide 3 Signals give priority to pedestrians to cross first and	4 A mix of activities and services activate the street from morning to night, making it safer and more interesting to walk 5 Street vendors and sidewalk amenities such as seating, shade, lighting and garbage bins attract more users and	6 7 Transit, such as bike share, bus, BRT, and rail, are reachable by foot 8 Small street widths are easier to cross 9 Slower speeds for traffic, by both design and enforced

Feet as **dynamic interface**
between the human body
and urban ground

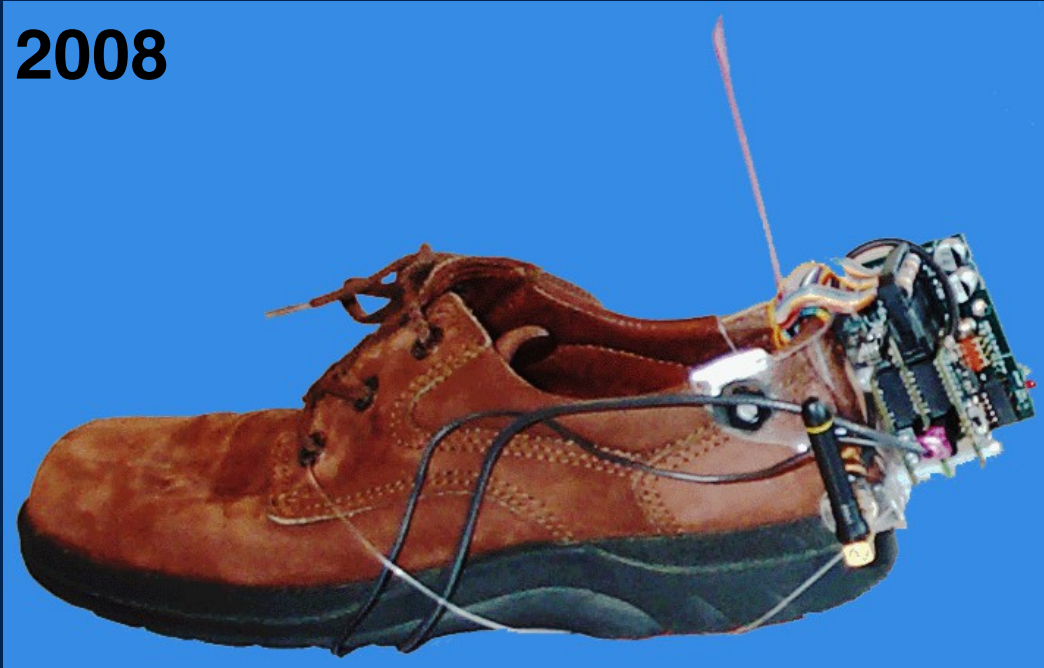


What if **just by walking** we could better understand

- how **ground conditions change** for urban walkers
 - how they **impact human health and experience**
 - how this information can **inform the design of urban spaces**
- 
- An abstract graphic on the right side of the slide. It features a light blue outline of a foot, with the heel and arch area filled with a darker blue. Below the foot, there are two wavy lines: a light blue one and a white one, suggesting ground or water levels.

Wearable Insole Pressure Sensors

2008



Capacitive Pressure Sensors. All other components are external

2026



Inductive Fabric Pressure Sensors Integrated Battery, IMU, Wireless Charging, Bluetooth.

Wearable Insole Pressure Sensors

100Hz sampling

12 pressure zones
(resistive fabric)

Bluetooth

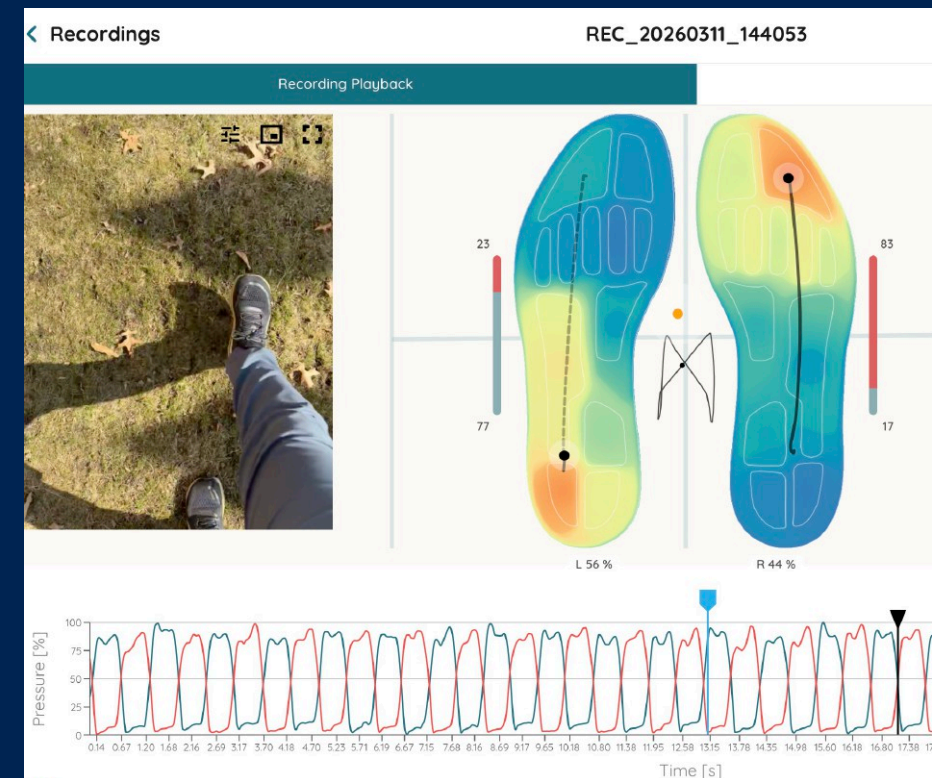
IMU (accelerometer,
gyroscope,
magnetometer)

Wireless
Charging
Battery

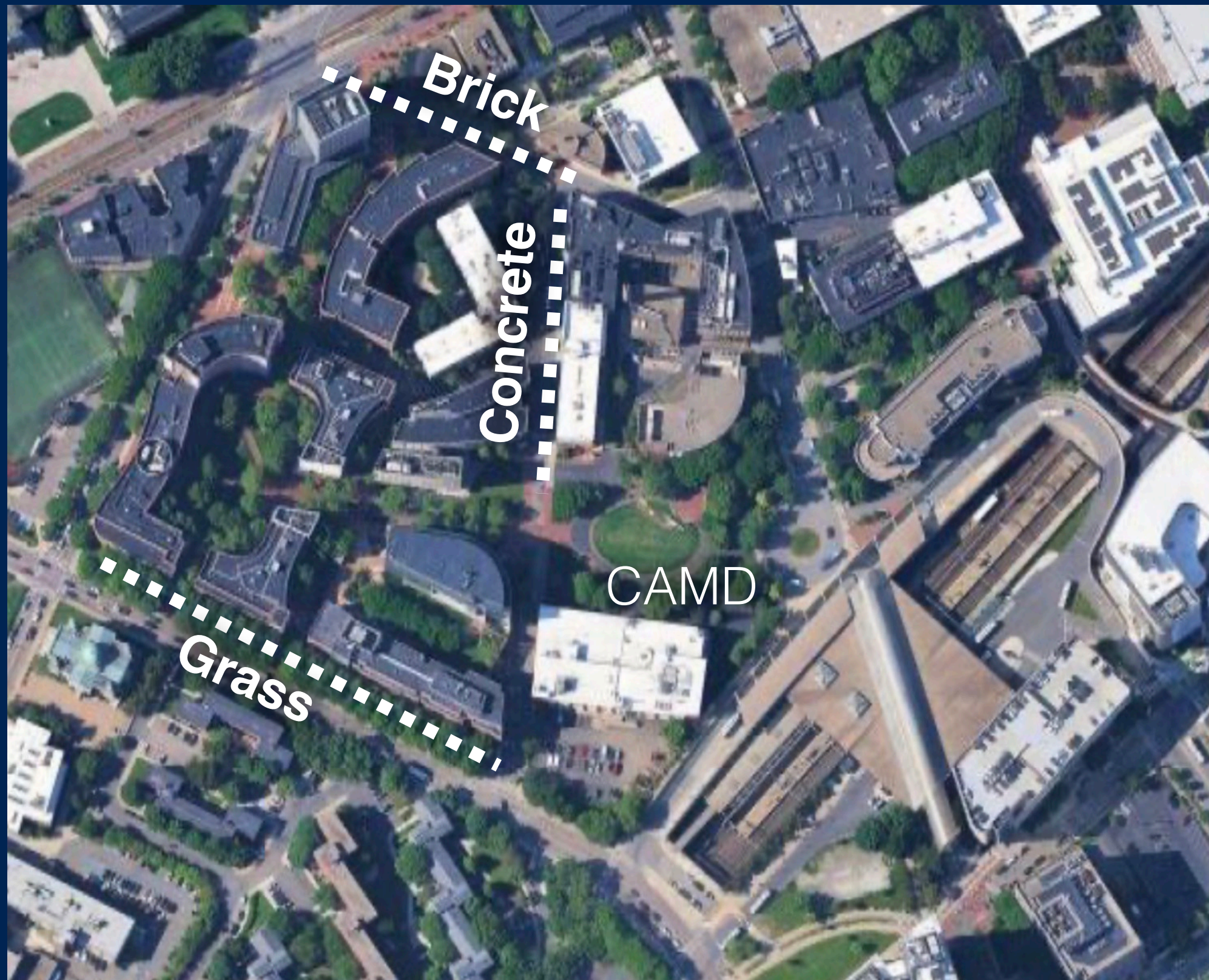


Technology Partner: Stappone

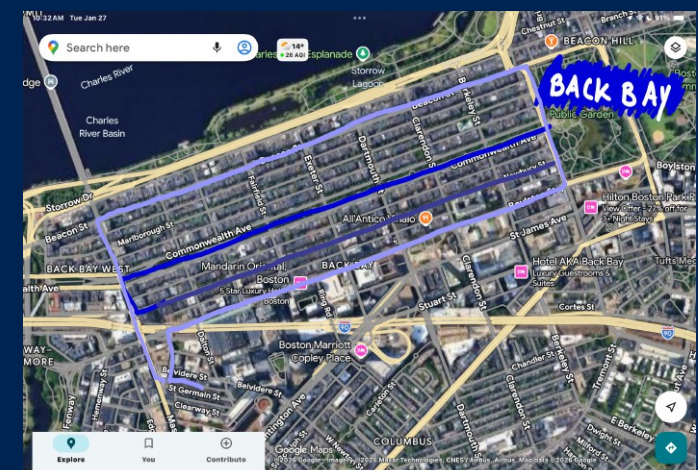
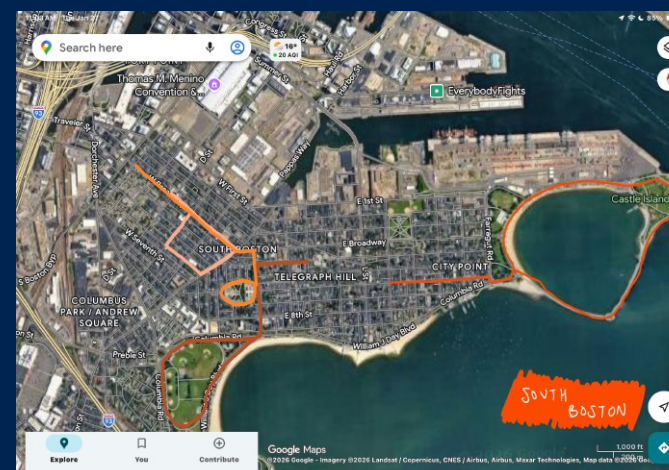
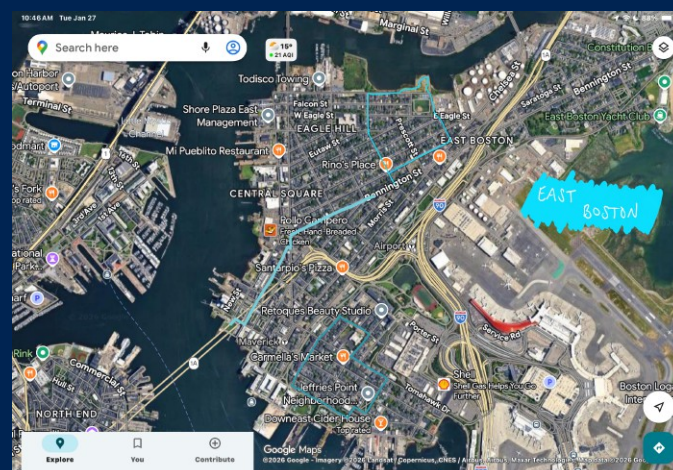
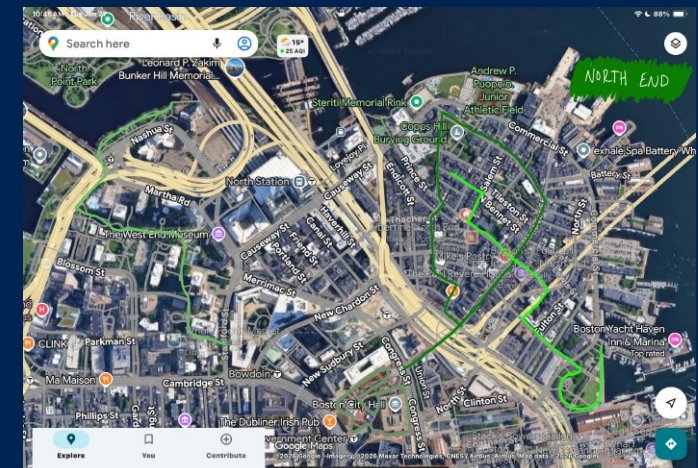
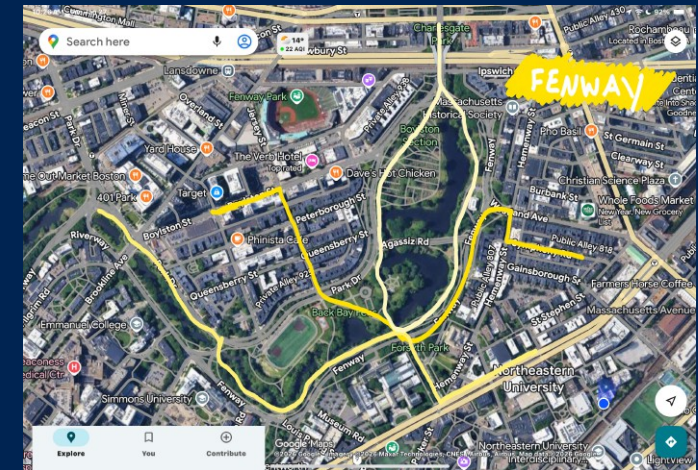
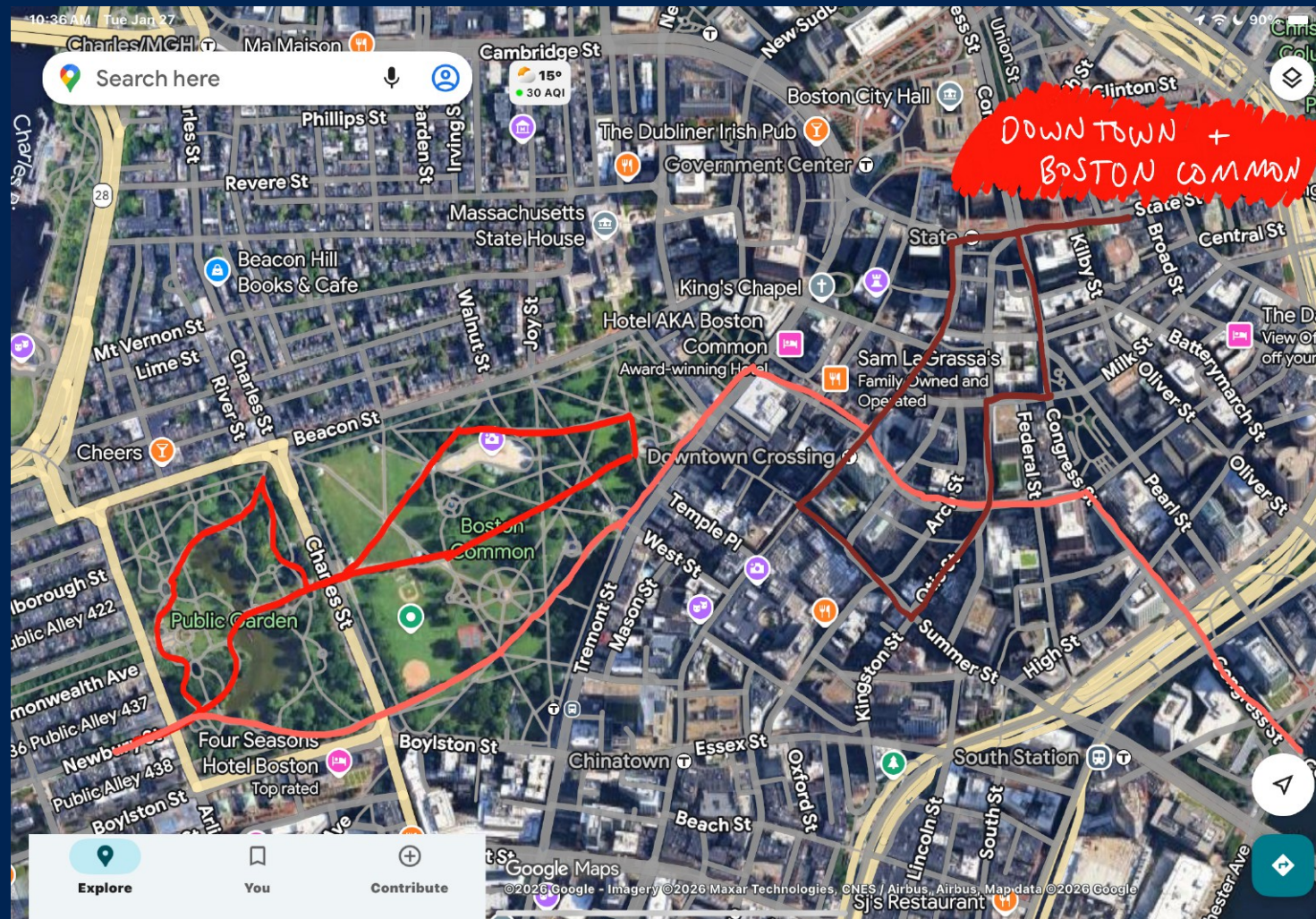
Data Collection while walking on different ground conditions



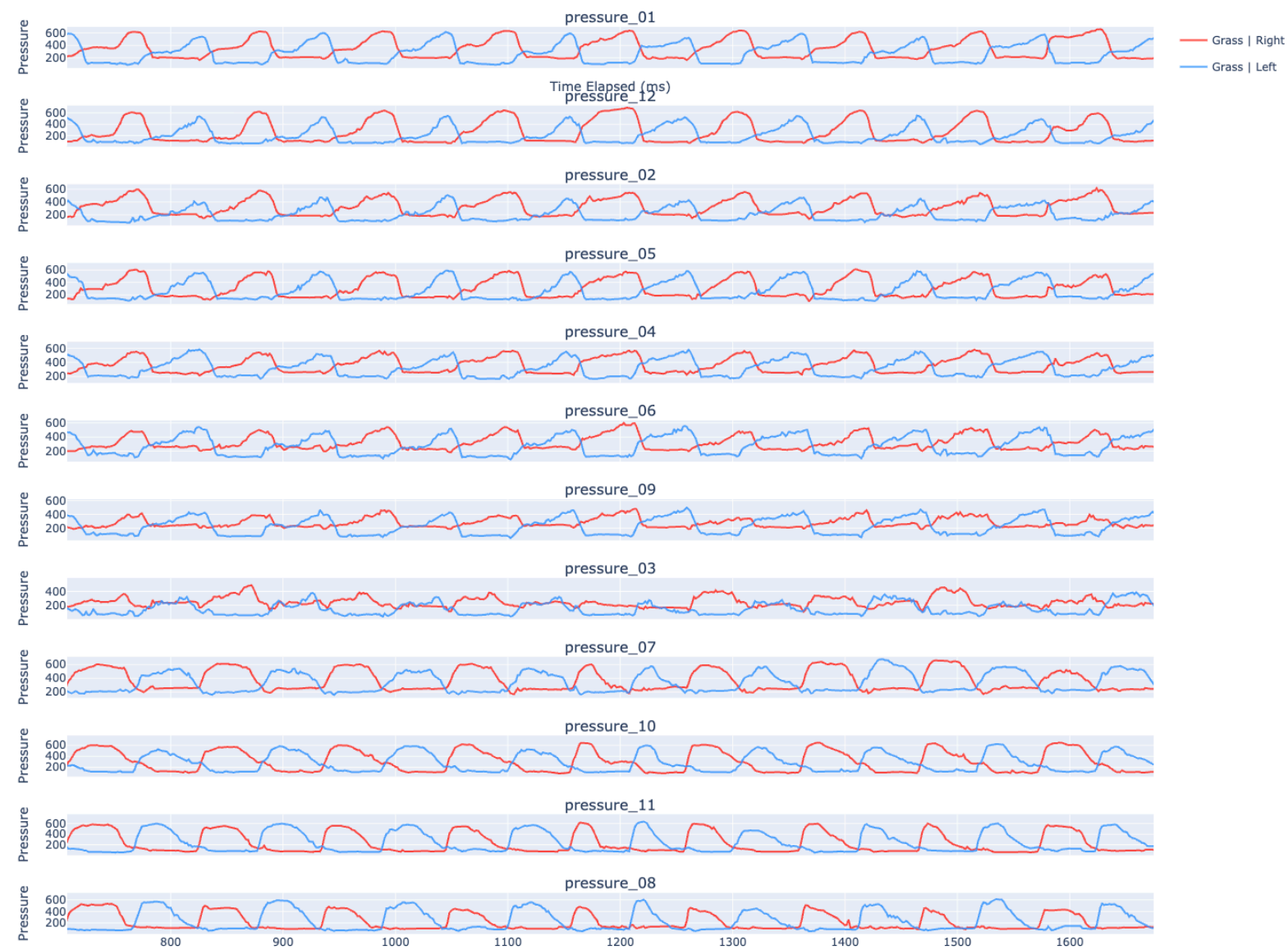
Test routes for diverse ground conditions



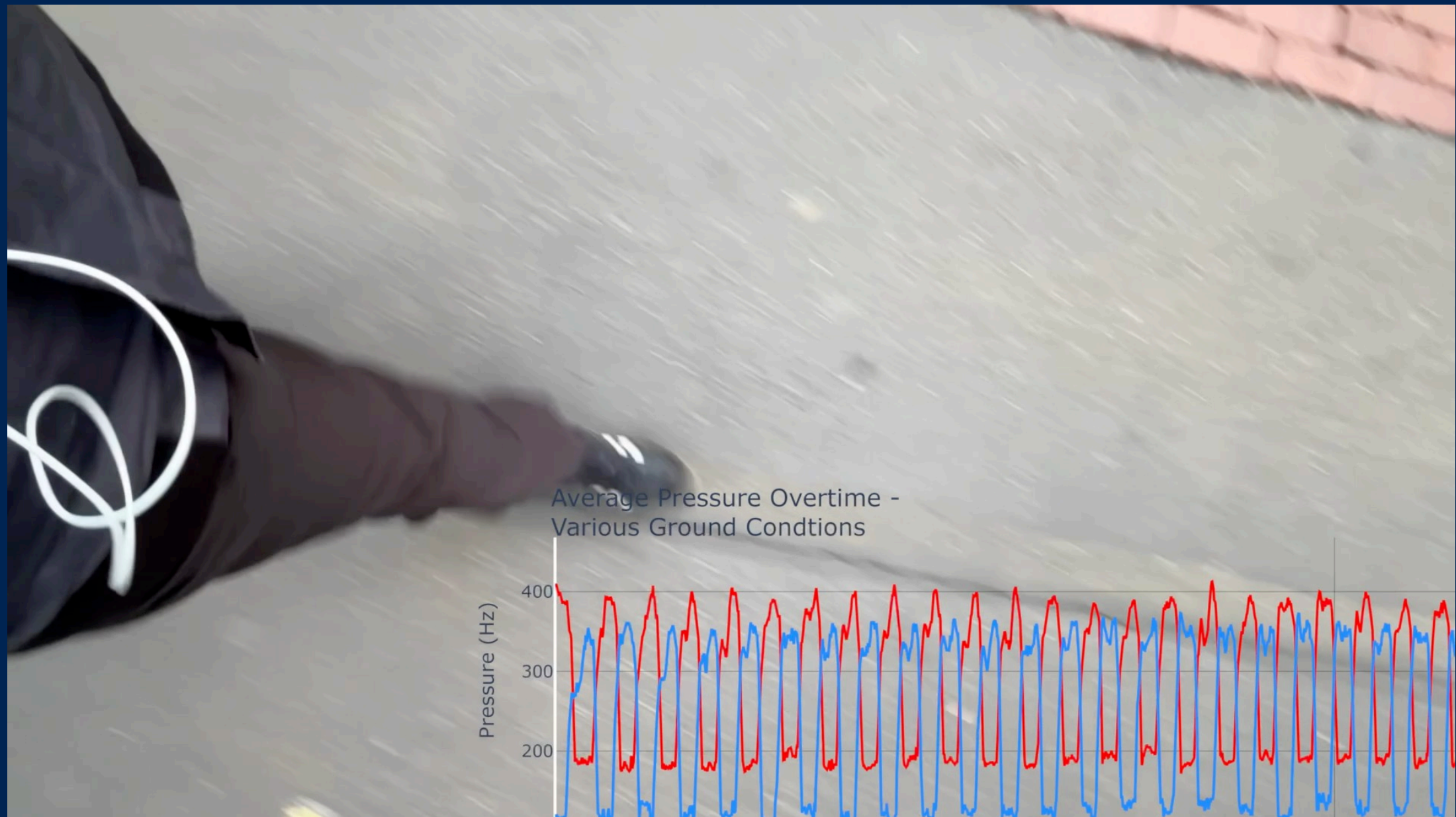
Interviews with urban walkers about walking routes



Data visualizations | Gait Cycle Time Series

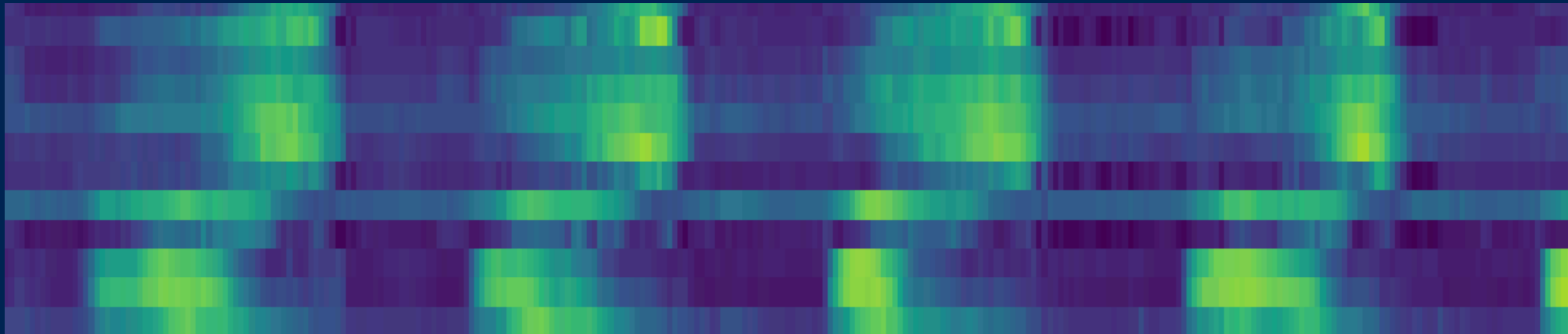


Data visualizations | Gait Cycle Time Series

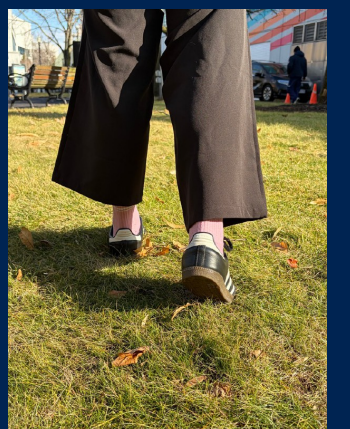
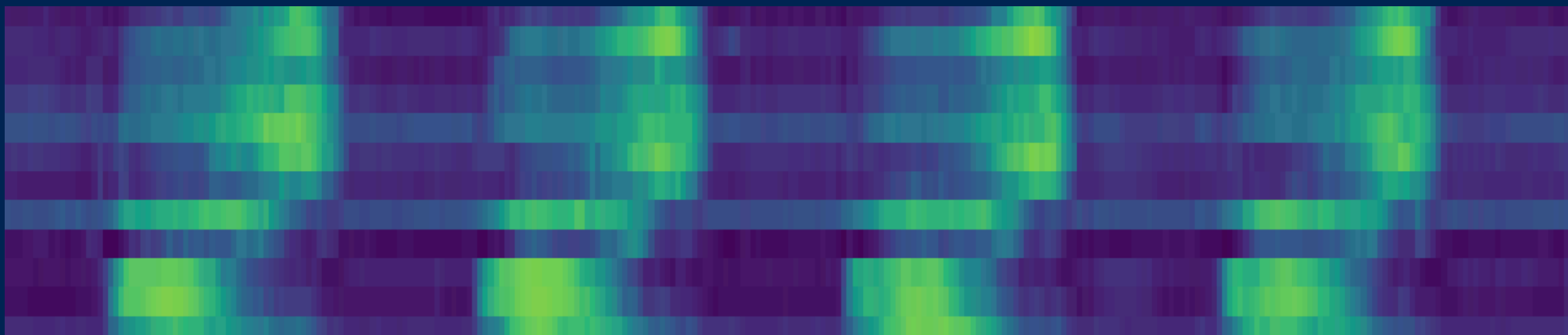


Data visualizations | Pressure Sensor Heatmaps

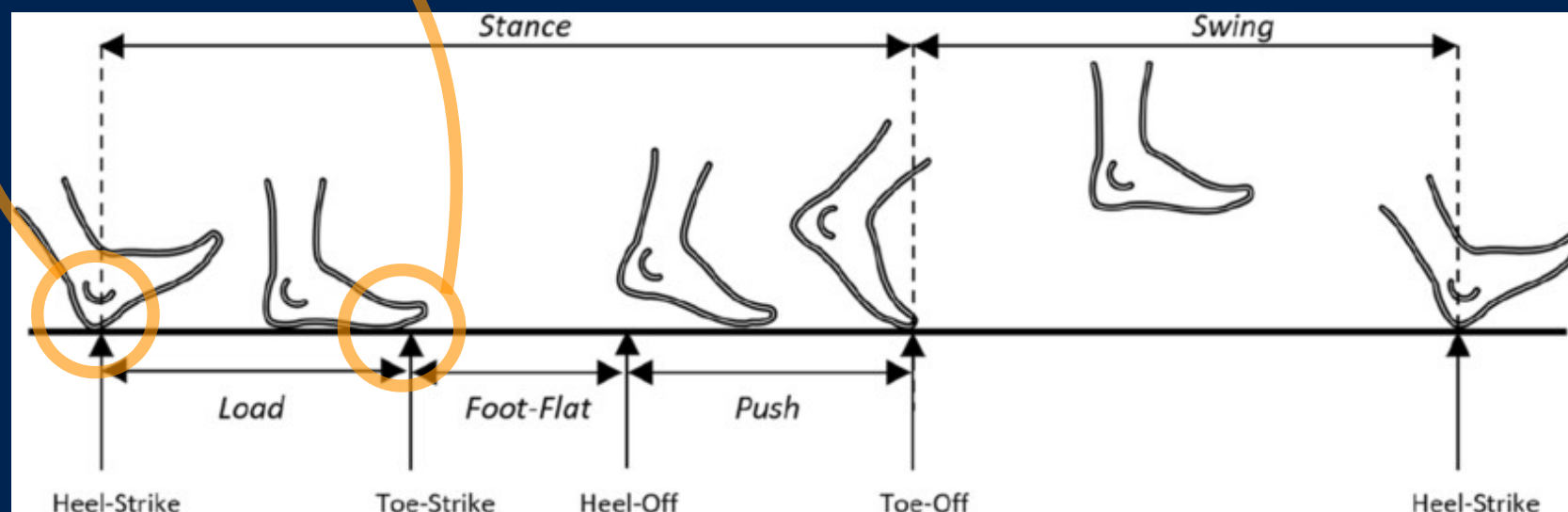
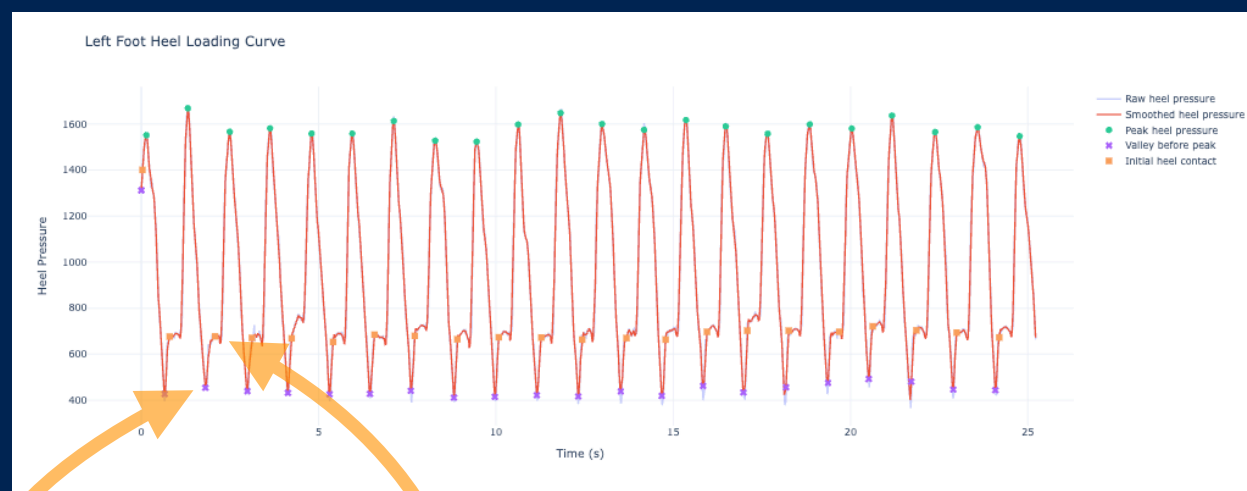
Asphalt



Grass



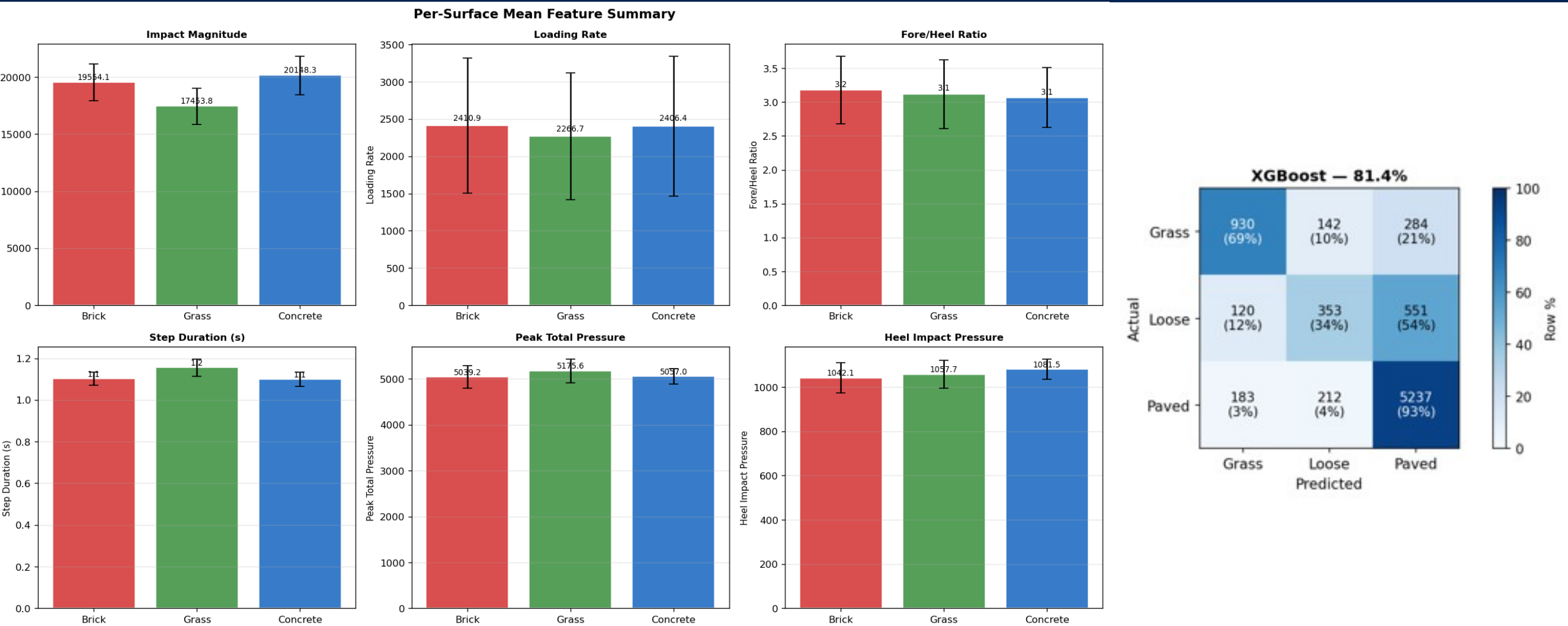
Data segmentation / Features / Machine learning



Feature Constructing at nexus of Physiotherapy, Urban Design and Public Health:

- Accelerometer Impact Peak at Heel Strike
- Spectral Energy Ratio (High vs Low Frequency)
- Loading Rate
- Pressure Distribution Entropy During Mid-Stance
- Step-to-Step Variability of Feature
- ...

Data segmentation / Features / Machine learning

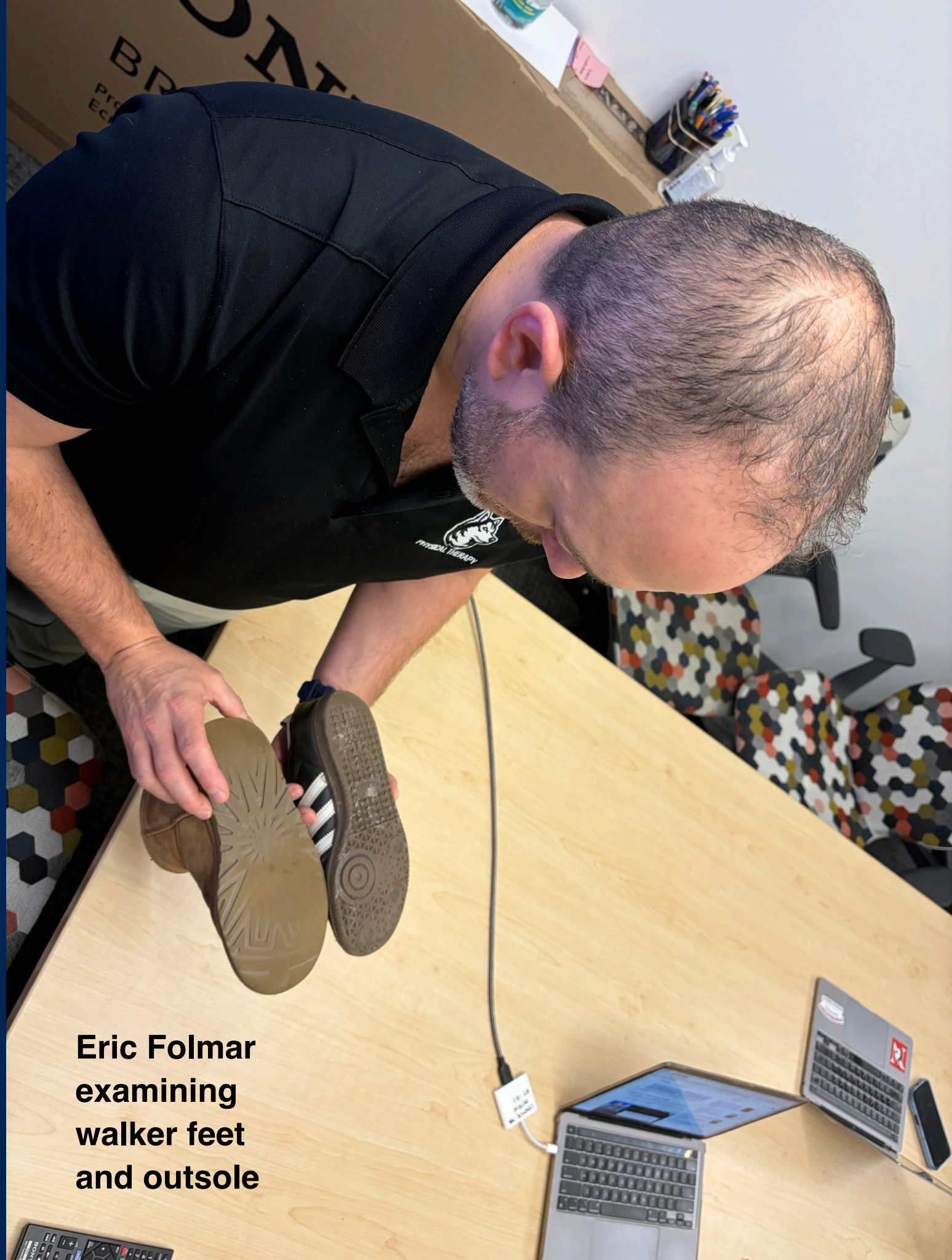


Data Traces

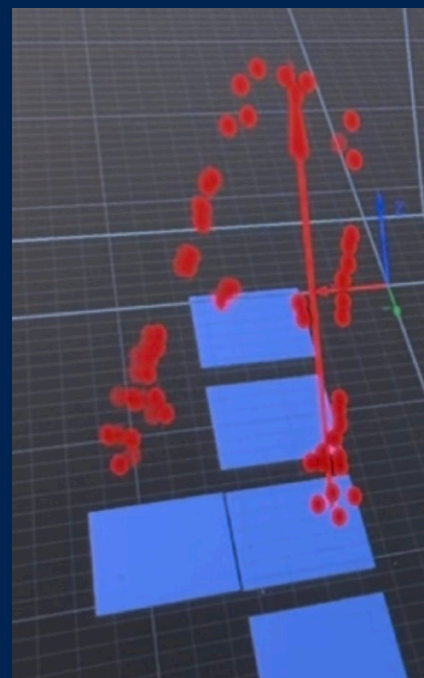
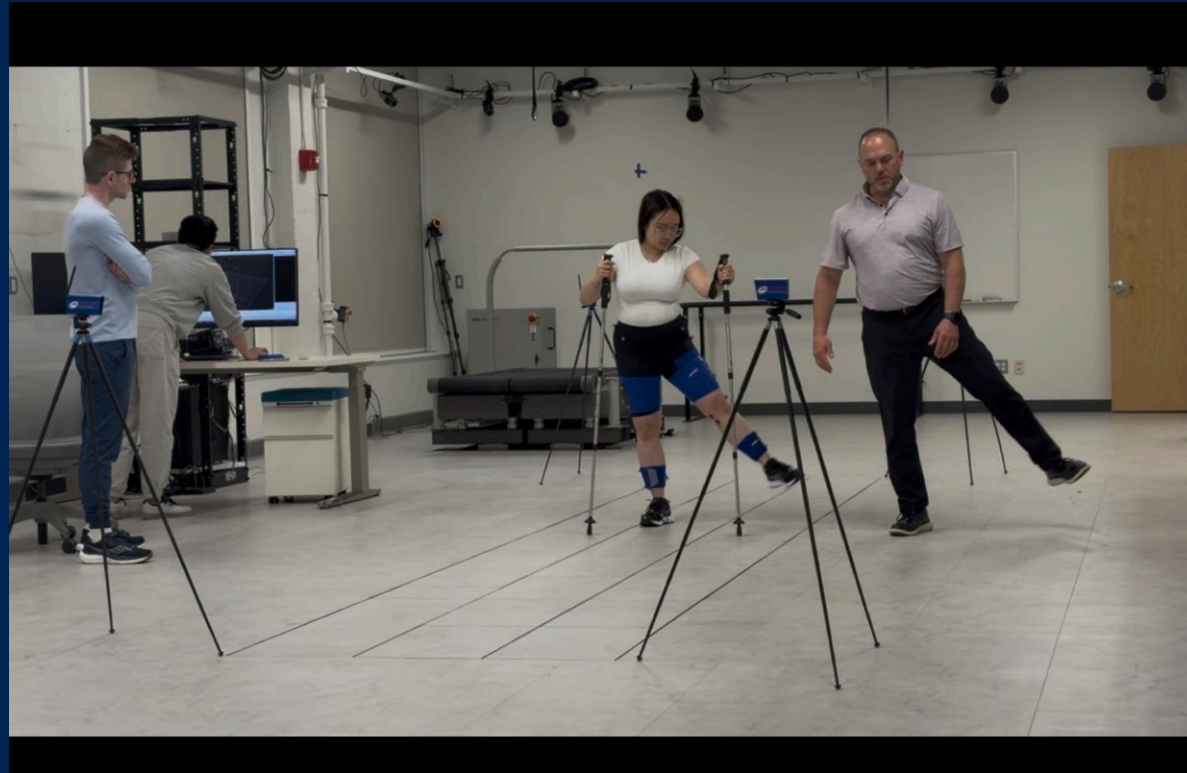
Foot and
biomechanics
assessment as
ground truth

Data traces mark
shoe soles based
on walking type

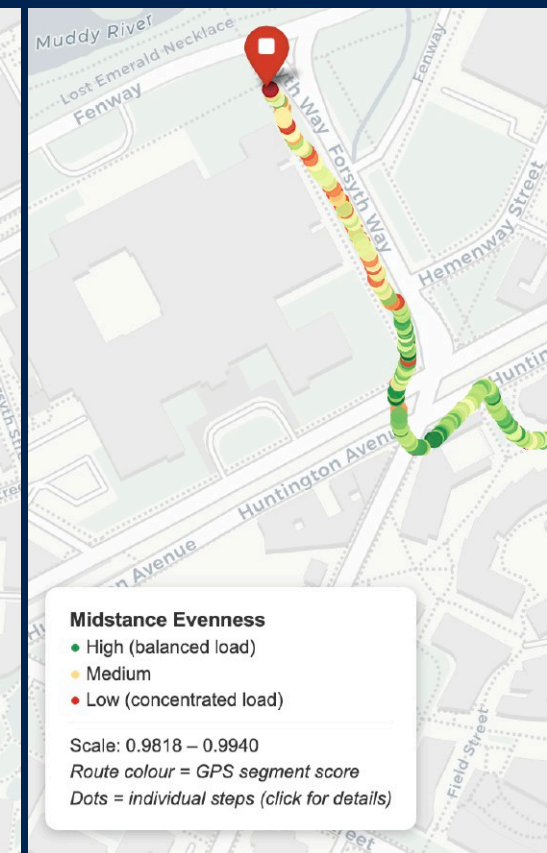
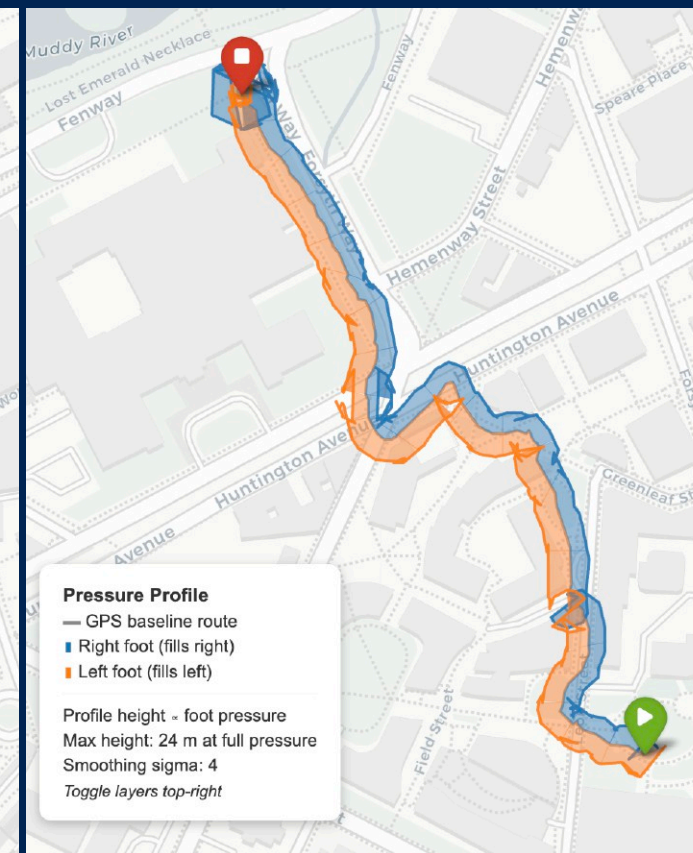
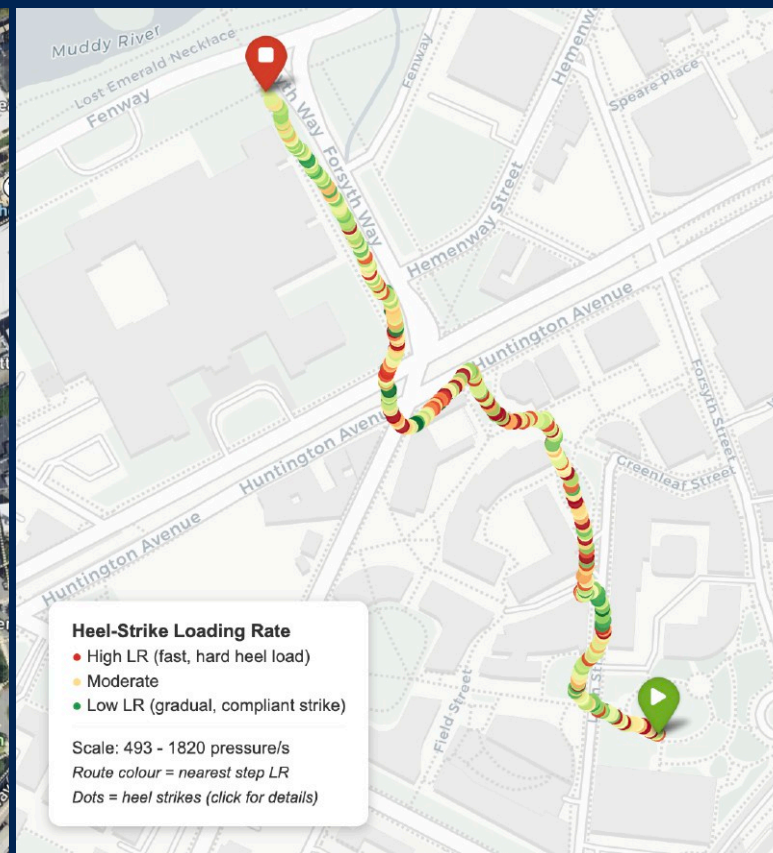
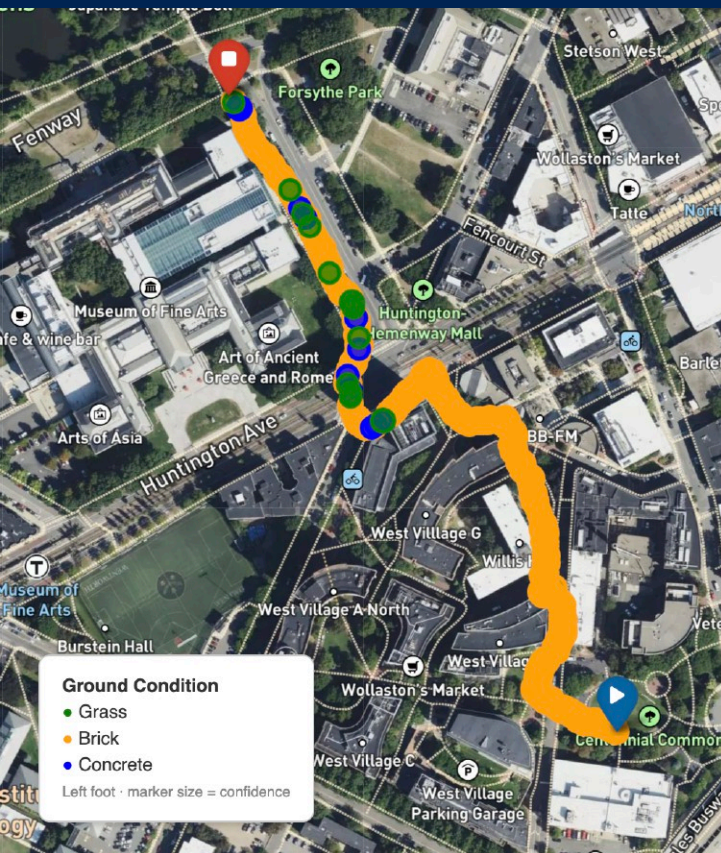
**Eric Folmar
examining
walker feet
and outsole**



Biomechanics Lab Data Collection



Data segmentation / Features / Machine learning



Next

From number of steps to **quality of steps**.

Map different **neighborhoods** by what **ground conditions** they afford for walking.

Develop framework to integrate this research in **design of walkable cities**.

How **different grounds** reinforce or complement a person's **walking type**.

Outreach:

- Pedestrian advocacy/research organization
- Shoe manufacturer



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